

Shubham Kumar

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Backend Engineer specializing in scalable distributed systems, event-driven architectures, and high-throughput, low-latency platforms. Expert in designing complex backend infrastructures and leading engineering teams to achieve high system reliability and performance at scale

Skills

- Languages & Backend: Python, SQL, Django, Django REST Framework, FastAPI, Microservices, WebSockets
- Distributed Systems: Kafka, Celery, Event-Driven Architecture, Asynchronous Processing, Redis
- Cloud & Infrastructure: AWS, Docker, Kubernetes, Nginx, Terraform, Prometheus, Grafana, OpenTelemetry
- AI & GenAI: LLMs, RAG, AI Agents, LangChain, LangGraph, pgvector (Vector Databases)
- Systems & Security: System Design, Concurrency, Reliability Engineering, Linux Memory Debugging, RBAC

Work Experience

Senior Backend Engineer – Collective Artists Network, Bangalore, India

Sep 2024 – Mar 2026

- Architected a personalized recommendation engine for **2M+ users** using **semantic search** and **community detection**, significantly increasing content discovery and user engagement
- Scaled platform engagement to **50K+ DAU** by designing a **high-concurrency, activity-driven leaderboard** system to drive user stickiness
- Integrated **OpenAI LLMs** for conversational profile setup, significantly reducing user onboarding drop-off rates.
- Spearheaded engineering excellence by establishing **observability and reliability standards**, reducing production incidents by **25%**

Senior Software Engineer – Radisys, Bangalore, India

Dec 2022 – Nov 2023

- Optimized system performance by reducing boot time by **60%** through a fault-tolerant supervision framework with automated recovery
- Resolved critical production crashes via deep-dive **core dump analysis and Linux memory debugging**, ensuring high system stability
- **Led a team of 5 engineers**, driving architectural decisions, enforcing code quality standards, and institutionalizing best practices

Software Engineer – Radisys, Bangalore, India

Jul 2021 – Dec 2022

- Automated system monitoring by developing a health monitor with remote log offloading to prevent storage capacity issues
- Enhanced platform reliability by boosting code **coverage to >90%** through robust testing of system monitoring modules

Projects

Mirai Botnet Reverse Engineering

- Reverse-engineered the Mirai botnet to analyze distributed attack coordination, propagation behavior, and C2 communication mechanisms.
- Simulated distributed botnet behavior across 100+ nodes to study DDoS traffic patterns and network saturation under load.
- Derived threat-detection insights by analyzing adversarial concurrency and distributed communication behavior.

Achievements

- Research: Published 3 peer-reviewed papers on IoT malware and distributed security in **IEEE and Springer**.
- Security: Competed in **100+ CTF competitions** (Web Security, Reverse Engineering, Linux Internals).
- Innovation: Developed and shipped **10+ functional prototypes** under aggressive hackathon timelines.

Education

NITK Surathkal – M.Tech(Research), Computer Science & Engineering – Information Security

Dec 2022

Lovely Professional University – B.Tech, Computer Science & Engineering

May 2019